Covariance Estimators and Out-of-Sample Hedging of SOFR Swap Risk*

Samson Lam

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1 Introduction

In interest rate derivatives, principal component analysis (PCA) hedging neutralizes a portfolio's exposure to the few factors that drive most yield-curve movements. Litterman and Scheinkman (1991) found that three factors, interpreted as level, slope and curvature, explain most of the variation in US Treasury returns, so a hedge against the leading factors needs only as many swaps as factors. The hedge and the risk that remains after it are both computed from an estimated covariance matrix of daily rate changes. This study examines whether the choice of covariance estimator affects the hedge effectiveness of a Secured Overnight Financing Rate (SOFR) swap portfolio and the accuracy of its out-of-sample risk forecasts, using daily rates from 14 swap tenors spanning 2024 to 2025.

The sample covariance matrix weights a year of changes equally. Ledoit and Wolf (2004) shrink it toward a scaled identity matrix with the shrinkage intensity that is asymptotically optimal under quadratic loss, and Chen et al. (2010) derive the oracle approximating shrinkage (OAS) intensity for Gaussian samples. An exponentially weighted moving average (EWMA) instead discounts older changes, and the RiskMetrics methodology uses a decay factor of 0.94 for daily data (J.P. Morgan/Reuters, 1996).

The portfolio is hedged with one or two swaps, either by neutralizing the leading principal components or by minimizing variance, under five estimators: the equal-weighted sample covariance matrix, the two shrinkage estimators and EWMA at two decay factors. No estimator consistently outperforms the equal-weighted one in hedge effectiveness. Shrinkage leaves PCA hedges unchanged and significantly worsens the minimum-variance hedge with two swaps on up to 38 of the 79 pairs, whereas EWMA significantly improves the PCA hedge with two swaps on 36 or 37 pairs and worsens it on 6 to 10. The estimator matters more for the risk forecast. As volatility falls in 2025, the equal-weighted estimator overstates the risk of the unhedged portfolio, EWMA with a decay factor of 0.94 shows no significant bias, and shrinkage overstates the variance of hedged portfolios by a factor of 2.5 to 4.

The remainder of the report is organized as follows. Section 2 describes the data. Section 3 states the hedges, the estimators and the evaluation. Section 4 reports the results, and Section 5 concludes.[1]

2 Data

Daily SOFR swap rates at the 14 tenors of 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25 and 30 years are obtained from Bloomberg for 2 January 2024 to 31 December 2025. Two holidays with partial quotes, 19 February and 4 July 2024, are removed, and daily changes are taken in basis points (bp). The first principal component (PC1) of the changes explains 93% to 95% of their variance in every half-year, and its daily standard deviation falls from about 20 bp to 13.7 bp in the second half of 2025 (Table 1).

*This project builds on homework for Fordham University's QFGB 8934 course.

[1]Code and replication data are available at *GitHub*.

Half-year	PC1 standard deviation (bp)	PC1 share of variance (%)
2024 H1	21.4	95.3
2024 H2	19.5	93.0
2025 H1	20.5	93.7
2025 H2	13.7	93.2

Table 1: PC1 of daily SOFR swap rate changes by half-year.

3 Methodology

3.1 Hedges

Write $\Delta \boldsymbol{y}_t \in \mathbb{R}^p$ ($p = 14$) for the vector of daily rate changes on day t, and $\boldsymbol{d}$ for the dollar value of 1 bp (DV01) of the portfolio by tenor, in dollars per bp. The DV01 profile is

$$\boldsymbol{d} = (-5.0, -2.0, -1.0, 3.5, -0.1, 1.0, 2.3, -0.8, -0.6, 2.5, 3.0, 4.0, 1.5, 5.0)^\top \times 10^3, \tag{1}$$

ordered from the 2-year to the 30-year tenor, with a net DV01 of \$13,300. The portfolio is hedged with $k \in \{1, 2\}$ swaps at the tenors selected by the matrix $B \in \mathbb{R}^{p \times k}$, whose columns are standard basis vectors of $\mathbb{R}^p$, and $\boldsymbol{h} \in \mathbb{R}^k$ holds their DV01s. The hedged DV01 and its profit and loss (P&L) on day t are

$$\boldsymbol{q} = \boldsymbol{d} + B\boldsymbol{h}, \qquad \pi_t = -\boldsymbol{q}^\top \Delta \boldsymbol{y}_t. \tag{2}$$

Let $\hat{\Sigma}$ be a covariance estimate with eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ ordered by decreasing eigenvalue, and $V_k = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$. The PCA hedge sets the exposure of $\boldsymbol{q}$ to the k leading components to zero, and the minimum-variance hedge minimizes $\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q}$,

$$\boldsymbol{h}_{\mathrm{PCA}} = -(V_k^\top B)^{-1} V_k^\top \boldsymbol{d}, \qquad \boldsymbol{h}_{\mathrm{MV}} = -(B^\top \hat{\Sigma} B)^{-1} B^\top \hat{\Sigma} \boldsymbol{d}. \tag{3}$$

The hedge sets are the 14 single tenors and the 79 of 91 pairs whose condition number $\kappa(V_2^\top B)$ is below 10 in the first estimation window, since an ill-conditioned pair turns small errors in V_2 into large hedges.

3.2 Covariance Estimators

The equal-weighted estimator is the sample covariance matrix S of the 252 daily changes before day t. The estimator of Ledoit and Wolf (2004), hereafter LW, and the OAS estimator of Chen et al. (2010) shrink S toward a scaled identity matrix,

$$\hat{\Sigma} = (1 - \delta) S + \delta \tau I, \qquad \tau = \mathrm{tr}(S)/p, \tag{4}$$

with the shrinkage intensity δ of each estimator given in Appendix A. The EWMA estimator applies the covariance recursion of J.P. Morgan/Reuters (1996) to all changes before day t, with the weights normalized to sum to one,

$$\hat{\Sigma} = \sum_{j \geq 1} w_j \, \Delta \boldsymbol{y}_{t-j} \Delta \boldsymbol{y}_{t-j}^\top, \qquad w_j \propto \lambda^{j-1}. \tag{5}$$

The decay factor λ is 0.94, the RiskMetrics value for daily data, or 0.98906, which corresponds to a half-life of 63 trading days.

3.3 Evaluation

On every test day t from 3 January to 31 December 2025, the portfolio is hedged with the estimates from the changes before day t, and the P&L π_t of (2) is recorded. Hedge effectiveness is measured by the residual risk

$$\mathrm{RR} = \frac{\sum_t \pi_{t,\mathrm{hedged}}^2}{\sum_t \pi_{t,\mathrm{unhedged}}^2}, \tag{6}$$

reported as the median over the single tenors or over the pairs. For each hedge, the squared P&L under each estimator is compared with that under the equal-weighted estimator by the test of Diebold and Mariano (1995), and the hedges significantly better or worse at the 5% level are counted. The risk forecast is evaluated by

$$z_t = \frac{\pi_t}{\sqrt{\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q}}}, \qquad \mathrm{H}_0\colon\ \mathbb{E}\left[z_t^2\right] = 1, \tag{7}$$

and the mean $\overline{z^2}$ of z_t^2 over the test days falls below one when the estimator overstates the risk. All tests use the t-statistic t_{NW} with the variance estimator of Newey and West (1987), stated in Appendix A.

4 Results

4.1 Hedge Effectiveness

In the median, a single swap removes 75% to 77% of the squared P&L and two swaps remove 97% to 98%, whichever estimator is used (Table 2). With one swap, the two rules leave almost the same residual risk, because PC1 accounts for most of the variance, and no estimator significantly improves any hedge (Table 3). With two swaps, the estimators separate: EWMA lowers the median residual risk of the PCA hedge from 2.56% to 2.32% and 2.34%, whereas shrinkage raises that of the minimum-variance hedge from 2.16% to 2.93% under LW and 2.69% under OAS. Shrinkage leaves the PCA hedges unchanged, as Proposition 1(i) below proves.

Rule	Swaps	Equal	LW	OAS	EWMA 0.94	EWMA 0.98906
PCA	1	22.9	22.9	22.9	24.8	23.2
	2	2.56	2.56	2.56	2.32	2.34
Minimum variance	1	23.0	23.0	23.0	24.7	23.3
	2	2.16	2.93	2.69	2.18	2.09

Table 2: Out-of-sample residual risk (%), median over 14 single tenors or 79 pairs.

Hedge by hedge, shrinkage significantly worsens the minimum-variance hedge with two swaps on 38 of the 79 pairs under LW and 30 under OAS, and improves none (Table 3). Substituting (4) into $\boldsymbol{h}_{\mathrm{MV}}$ of (3) yields

$$B^\top \hat{\Sigma} B = (1-\delta) B^\top S B + \delta\tau I_k, \qquad B^\top \hat{\Sigma} \boldsymbol{d} = (1-\delta) B^\top S \boldsymbol{d} + \delta\tau \boldsymbol{d}_B, \tag{8}$$

where I_k is the $k \times k$ identity matrix and $\boldsymbol{d}_B = B^\top \boldsymbol{d}$ is the DV01 at the hedge tenors. The shrinkage target alone ($\delta = 1$) would yield $\boldsymbol{h}_{\mathrm{MV}} = -\boldsymbol{d}_B$, a hedge that offsets only the DV01 at the hedge tenors and ignores the correlation across the curve, and shrinkage pulls the hedge toward it. For two swaps whose rates move almost together, the smaller eigenvalue of $B^\top S B$ is small, so adding $\delta\tau$ to it changes the hedge materially. EWMA matters for the PCA hedges instead. With two swaps, decay factors of 0.94 and 0.98906 significantly improve 37 and 36 of the 79 pairs and worsen 10 and 6, while the minimum-variance hedges differ significantly from the equal-weighted ones on at most 1 pair. With one swap, a decay factor of 0.94 worsens 4 of the 14 hedges under both rules.

			PCA		Minimum variance	
Swaps	Estimator	Hedges	Better	Worse	Better	Worse
1	LW	14	–	–	0	0
	OAS	14	–	–	0	0
	EWMA 0.94	14	0	4	0	4
	EWMA 0.98906	14	0	0	0	0
2	LW	79	–	–	0	38
	OAS	79	–	–	0	30
	EWMA 0.94	79	37	10	0	0
	EWMA 0.98906	79	36	6	1	0

Table 3: Hedges with significantly lower (better) or higher (worse) mean squared P&L than under the equal-weighted estimator, by the test of Diebold and Mariano (1995) at the 5% level. Under LW and OAS, PCA hedges equal those of the equal-weighted estimator and are not tested.

4.2 Risk Forecasts

For the unhedged portfolio, the equal-weighted, LW and OAS forecasts significantly overstate the risk, with $\overline{z^2}$ of 0.732 to 0.737 (Table 4). Their 252-day window still carries the higher volatility of the preceding year when the volatility of PC1 falls in 2025 (Table 1). EWMA with a decay factor of 0.94 tracks this decline and shows no significant bias, whereas a decay factor of 0.98906 removes little of the bias ($\overline{z^2} = 0.770$).

For the hedged portfolios, the ranking changes. The equal-weighted forecast shows no significant bias, the realized variance under EWMA with a decay factor of 0.94 exceeds the forecast by 24% to 25% without being significant at the 5% level, and shrinkage overstates the variance by a factor of about 2.5 to 4.

	Unhedged		PCA, 2 swaps		Minimum variance, 2 swaps	
Estimator	$\overline{z^2}$	t_{NW}	$\overline{z^2}$	t_{NW}	$\overline{z^2}$	t_{NW}
Equal	0.732	−3.24	1.03	0.22	0.964	−0.30
LW	0.737	−3.17	0.237	−24.7	0.388	−9.05
OAS	0.735	−3.19	0.296	−18.4	0.408	−9.23
EWMA 0.94	0.961	−0.44	1.24	1.70	1.25	1.73
EWMA 0.98906	0.770	−2.83	0.984	−0.13	0.948	−0.44

Table 4: Calibration of the risk forecast, with t_{NW} for $\mathrm{H}_0\colon\ \mathbb{E}[z_t^2] = 1$. For hedged portfolios, z_t^2 is averaged over the 79 pairs on each day.

Proposition 1 states both effects of shrinkage, and Appendix B contains its proof.

Proposition 1. *Let $\hat{\Sigma} = (1-\delta)S + \delta\tau I$ with $0 \le \delta < 1$ and $\tau = \mathrm{tr}(S)/p$, and let S have distinct eigenvalues. Then*

(i) $\hat{\Sigma}$ and S have the same eigenvectors in the same order, so the PCA hedge is the same under both, and

(ii) for every $\boldsymbol{q}$ with $\boldsymbol{q}^\top S\boldsymbol{q} > 0$,

$$\textit{Inflation} = \frac{\boldsymbol{q}^\top \hat{\Sigma}\boldsymbol{q}}{\boldsymbol{q}^\top S\boldsymbol{q}} = (1-\delta) + \frac{\delta\tau}{\rho(\boldsymbol{q})}, \qquad \rho(\boldsymbol{q}) = \frac{\boldsymbol{q}^\top S\boldsymbol{q}}{\boldsymbol{q}^\top\boldsymbol{q}}. \tag{9}$$

Part (i) holds because shrinkage maps every eigenvalue ℓ to $(1-\delta)\ell+\delta\tau$, an increasing map toward their grand mean τ (Ledoit and Wolf, 2004), and leaves the eigenvectors unchanged. By part (ii), the inflation of the forecast variance depends on the Rayleigh quotient $\rho(\boldsymbol{q})$, the variance of the portfolio per unit of squared DV01. An unhedged portfolio is dominated by the level factor and has a large $\rho(\boldsymbol{q})$, so the inflation is negligible. A hedged portfolio has its exposure to the leading components removed and a small $\rho(\boldsymbol{q})$, so the term $\delta\tau/\rho(\boldsymbol{q})$ inflates its forecast substantially, as confirmed by Table 5.

		Inflation	
Portfolio	$\rho(\boldsymbol{q})$ (bp^2)	LW	OAS
Unhedged	52.7	0.994	0.996
PCA, 2 swaps	0.108	4.47	3.43

Table 5: Median Rayleigh quotient and median inflation over the test days, and for PCA hedges also over the 79 pairs.

5 Conclusion

This study examines whether the choice of covariance estimator affects the out-of-sample hedge effectiveness and risk forecasts of a SOFR swap portfolio. A DV01 profile over 14 tenors is hedged with one or two swaps by PCA or minimum-variance hedging under five estimators. Each hedge is evaluated over 2025 by Diebold–Mariano tests on its squared P&L and by a calibration test of its risk forecast.

Hedge effectiveness depends little on the estimator, since the median hedge with two swaps removes 97% to 98% of the squared P&L under every estimator. Shrinkage leaves the PCA hedges unchanged and pulls the minimum-variance hedge toward offsetting only the DV01 at the hedge tenors, which worsens it on up to 38 of the 79 pairs, while EWMA improves the PCA hedge with two swaps on 36 or 37 pairs.

The risk forecast depends strongly on the estimator. The equal-weighted estimator lags the fall in volatility and overstates the unhedged risk, EWMA with a decay factor of 0.94 shows no significant bias, and shrinkage overstates the variance of hedged portfolios by a factor of 2.5 to 4, because the term $\delta\tau/\rho(\boldsymbol{q})$ of Proposition 1 is large when the Rayleigh quotient of a hedged portfolio is small.

These results rest on a single portfolio over 2025, a year of falling volatility, so the ranking of the estimators for the risk forecast may shift in high-volatility periods. Extending the sample across several economic cycles and other DV01 profiles would test whether the ranking holds.

References

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A Definitions

Write $\boldsymbol{x}_s$, $s = 1, \dots, W$, for the $W = 252$ demeaned changes in the estimation window, so that $S = W^{-1} \sum_s \boldsymbol{x}_s \boldsymbol{x}_s^\top$, and $\|A\|_F^2 = \operatorname{tr}(AA^\top)$. The LW intensity is the feasible shrinkage intensity of Ledoit and Wolf (2004), and the OAS intensity is that of Chen et al. (2010),

$$\delta_{\mathrm{LW}} = \frac{\min\left(\bar{b}^2, \|S - \tau I\|_F^2\right)}{\|S - \tau I\|_F^2}, \qquad \bar{b}^2 = \frac{1}{W^2} \sum_{s=1}^{W} \left\|\boldsymbol{x}_s \boldsymbol{x}_s^\top - S\right\|_F^2, \tag{10}$$

$$\delta_{\mathrm{OAS}} = \min\left(1, \frac{\operatorname{tr}(S^2) + \operatorname{tr}^2(S)}{(W+1)\left(\operatorname{tr}(S^2) - \operatorname{tr}^2(S)/p\right)}\right). \tag{11}$$

The condition number is $\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$, the ratio of the largest to the smallest singular value. For a daily series g_t, $t = 1, \dots, T$, with mean $\bar{g}$, the test statistic uses the variance estimator of Newey and West (1987),

$$t_{\mathrm{NW}} = \frac{\bar{g}}{\sqrt{\hat{\Omega}/T}}, \qquad \hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{l=1}^{L} \left(1 - \frac{l}{L+1}\right) \hat{\gamma}_l, \tag{12}$$

where $\hat{\gamma}_l$ is the sample autocovariance of g_t at lag l. The lag is set to $L = \lfloor 4(T/100)^{2/9} \rfloor$, which is 4 for $T = 251$. The test of Diebold and Mariano (1995) applies t_{NW} to the loss differential $g_t = \pi_{t,e}^2 - \pi_{t,\mathrm{Equal}}^2$ for estimator e, and the calibration test applies it to $g_t = z_t^2 - 1$. Under $\mathrm{H}_0 \colon \mathbb{E}[g_t] = 0$, t_{NW} is asymptotically standard normal, and a hedge is counted as better (worse) when $t_{\mathrm{NW}} < -1.96$ ($t_{\mathrm{NW}} > 1.96$).

B Proof of Proposition 1

Proof. Let $S\boldsymbol{v}_i = \ell_i \boldsymbol{v}_i$ with $\ell_1 > \cdots > \ell_p$. Then

$$\hat{\Sigma}\boldsymbol{v}_i = (1-\delta)\ell_i \boldsymbol{v}_i + \delta\tau \boldsymbol{v}_i = \big((1-\delta)\ell_i + \delta\tau\big)\boldsymbol{v}_i, \tag{13}$$

so $\boldsymbol{v}_i$ is an eigenvector of $\hat{\Sigma}$. Since $\delta < 1$, the map $\ell \mapsto (1-\delta)\ell + \delta\tau$ is strictly increasing and preserves eigenvalue order. Hence, V_k is the same under $\hat{\Sigma}$ and S up to the signs of its columns. Since $(DV_k^\top B)^{-1} D V_k^\top \boldsymbol{d} = (V_k^\top B)^{-1} V_k^\top \boldsymbol{d}$ for any $D = \operatorname{diag}(\pm 1)$, the signs do not change $\boldsymbol{h}_{\mathrm{PCA}}$ in (3), which depends on the estimate only through V_k. For part (ii),

$$\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q} = (1-\delta)\, \boldsymbol{q}^\top S \boldsymbol{q} + \delta\tau\, \boldsymbol{q}^\top \boldsymbol{q}, \tag{14}$$

and dividing by $\boldsymbol{q}^\top S \boldsymbol{q}$ yields (9). □